

Week 12

Markov's Inequality

Chebyshev's Inequality

Maximum Likelihood Estimation

Exponential Distribution

Bernoulli Bias

Binomial(m,p) Distribution

Poisson

Uniform Distribution

Normal Distribution

Markov's Inequality

$$P(x \geq t) \leq \frac{\mu}{t}$$

Chebyshev's Inequality

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2} \text{ or } \frac{\text{Var}(X)^2}{t^2}$$

$$P(X > \mu + k\sigma) + P(X < \mu - k\sigma) \leq \frac{1}{k^2}$$

Maximum Likelihood Estimation

Exponential Distribution

$$\hat{\lambda} = \frac{n}{\sum x_i}$$

Bernoulli Bias

$$\hat{\theta} = \frac{\sum x_i}{n}$$

Binomial(m,p) Distribution

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$$\hat{p} = \frac{\Sigma x_i}{np}$$

Poisson

$$\hat{\lambda} = \bar{x}$$

Uniform Distribution

$$\hat{a} = \min(x_1, x_2, x_2, ..x_n)$$

$$\hat{b} = \max(x_1, x_2, x_2, ..x_n)$$

Normal Distribution

$$\hat{\mu} = \frac{\Sigma x_i}{n}$$

$$\hat{\sigma} = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$